STUDY NOTES

CTET Paper II Mathematics (English)

Complete 200 Practice MCQs (Content + Pedagogy)

2 MCQ/source file(s)

27 August 2026

200 MCQs

विषय-सूची / Contents

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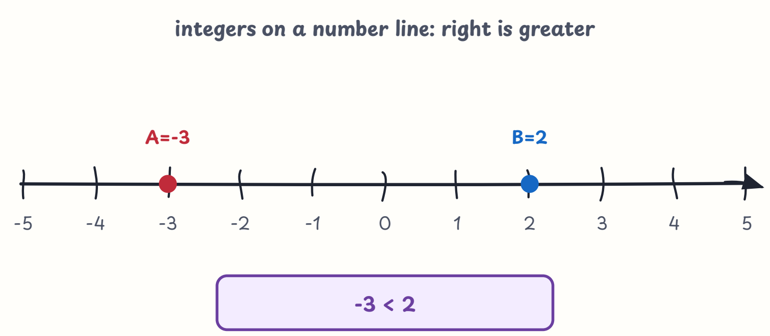
CTET Paper II Mathematics MCQ

Part 1: Mathematics Content — Q001-Q100

Set 01 — Number System and Integers (Q001-Q010)

Class VI-VIII number sense: whole numbers, negative numbers and operations with integers.



चित्र 1 — compare integer positions on a number line

Q001. **Which is the greatest number among −7, −2, 0 and −5?**

- A. 0
- B. −7
- C. −2
- D. −5

Answer: A

Explanation: On the number line, 0 lies to the right of all the given negative integers.

Q002. What is the absolute value of -8?

- A. -8
- B. 8
- C. 0
- D. 16

Answer: B

Explanation: Absolute value is the distance from zero, so $|-8| = 8$.

Q003. Which integer lies immediately to the left of -2 on a number line?

- A. -1
- B. 2
- C. -3
- D. 3

Answer: C

Explanation: Moving one step left decreases the integer by 1: $-2 - 1 = -3$.

Q004. In the figure, which comparison is correct for the marked integers?

- A. $-3 > 2$
- B. $-3 = 2$
- C. $2 < -3$
- D. $-3 < 2$

Answer: D

Explanation: The point at -3 is to the left of the point at 2, so -3 is smaller.

Q005. What is the additive inverse of -13?

- A. 13
- B. -13
- C. 0
- D. 26

Answer: A

Explanation: A number and its additive inverse have sum zero; $-13 + 13 = 0$.

Q006. Calculate: (−6) + 9

- A. −15
- B. 3
- C. −3
- D. 15

Answer: B

Explanation: Starting at −6 and moving 9 units to the right gives 3.

Q007. Calculate: 7 − 12

- A. 5
- B. 19
- C. −5
- D. −19

Answer: C

Explanation: Subtracting 12 from 7 gives 7 + (−12) = −5.

Q008. Calculate: (−4) × (−3)

- A. −12
- B. 7
- C. −7
- D. 12

Answer: D

Explanation: The product of two negative integers is positive, and 4 × 3 = 12.

Q009. Calculate: (−24) ÷ 6

- A. −4
- B. 4
- C. −30
- D. 30

Answer: A

Explanation: A negative divided by a positive is negative; 24 ÷ 6 = 4.

Q010. The temperature is −3°C and rises by 8°C. What is the new temperature?

- A. −11°C
- B. 5°C
- C. −5°C
- D. 11°C

Answer: B

Explanation: A rise of 8 means add 8: $-3 + 8 = 5$°C.

Set 02 — Factors, Multiples and Divisibility (Q011–Q020)

Q011. What is the HCF of 18 and 24?

- A. 2
- B. 3
- C. 6
- D. 12

Answer: C

Explanation: The common factors are 1, 2, 3 and 6; the greatest is 6.

Q012. What is the LCM of 8 and 12?

- A. 4
- B. 16
- C. 96
- D. 24

Answer: D

Explanation: Multiples meet first at 24: $8 \times 3 = 12 \times 2 = 24$.

Q013. Which of the following is a prime number?

- A. 29
- B. 21
- C. 27
- D. 35

Answer: A

Explanation: 29 has exactly two factors, 1 and 29.

Q014. Which is NOT a factor of 36?

- A. 2
- B. 5
- C. 3
- D. 6

Answer: B

Explanation: 36 is not divisible by 5, whereas it is divisible by 2, 3 and 6.

Q015. A number is divisible by 9 when—

- A. its last digit is 0 or 5
- B. its last two digits are divisible by 4
- C. the sum of its digits is divisible by 9
- D. it is an even number

Answer: C

Explanation: The divisibility test for 9 uses the sum of the digits.

Q016. What is the value of $2^3 \times 3$?

- A. 12
- B. 18
- C. 36
- D. 24

Answer: D

Explanation: $2^3 = 8$ and $8 \times 3 = 24$.

Q017. What is the greatest common factor of 18 and 30?

- A. 6
- B. 3
- C. 9
- D. 12

Answer: A

Explanation: Factors common to both include 1, 2, 3 and 6; 6 is greatest.

Q018. If two numbers are 8 and 12, their HCF is 4. What is their LCM?

- A. 16
- B. 24
- C. 20
- D. 96

Answer: B

Explanation: For positive integers, product = HCF × LCM: $8 \times 12 = 4 \times \text{LCM}$, so LCM = 24.

Q019. Which is the smallest prime number?

- A. 0
- B. 1
- C. 2
- D. 3

Answer: C

Explanation: 2 is prime and is the first positive integer with exactly two factors.

Q020. The number 1 is—

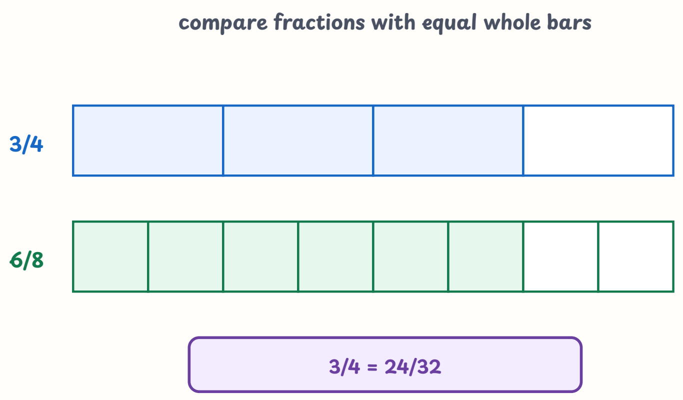
- A. prime only
- B. composite only
- C. both prime and composite
- D. neither prime nor composite

Answer: D

Explanation: 1 has only one factor, so it does not meet the definition of either class.

Set 03 — Fractions and Decimals (Q021-Q030)

> Fractions are numbers representing equal parts; visual models help learners compare size and equivalence.



चित्र 2 — *equivalent fractions shown with equal whole bars*

Q021. The two bars in the figure represent—

- A. equivalent fractions: 3/4 = 6/8
- B. unlike fractions: 3/4 < 6/8
- C. fractions with different wholes
- D. whole numbers 3 and 6

Answer: A

Explanation: Both bars show the same part of an equal-sized whole, so 3/4 and 6/8 are equivalent.

Q022. Calculate: 2/3 + 1/6

- A. 3/9
- B. 5/6
- C. 1/2
- D. 2/9

Answer: B

Explanation: Using denominator 6, 2/3 = 4/6; therefore 4/6 + 1/6 = 5/6.

Q023. Calculate: 5/8 − 1/4

- A. 4/8
- B. 1/8
- C. 3/8
- D. 5/4

Answer: C

Explanation: 1/4 = 2/8, so 5/8 − 2/8 = 3/8.

Q024. What is 3/5 × 10/9 in simplest form?

- A. 1/3
- B. 2/5
- C. 6/9
- D. 2/3

Answer: D

Explanation: Cancel 5 with 10 and 3 with 9: (1 × 2)/(1 × 3) = 2/3.

Q025. What is (4/7) ÷ (2/7)?

- A. 2
- B. 2/7
- C. 8/49
- D. 1/2

Answer: A

Explanation: Dividing by 2/7 means multiplying by 7/2: 4/7 × 7/2 = 2.

Q026. The decimal 0.75 is equal to—

- A. 1/4
- B. 3/4
- C. 7/5
- D. 75/10

Answer: B

Explanation: 0.75 = 75/100 = 3/4 after dividing numerator and denominator by 25.

Q027. **Which is the greatest: 0.7, 2/3, 0.65 or 5/8?**

- A. 2/3
- B. 0.65
- C. 0.7
- D. 5/8

Answer: C

Explanation: 2/3 is about 0.667 and 5/8 is 0.625, so 0.7 is greatest.

Q028. **Calculate: 1 1/2 + 2 3/4**

- A. 3 1/4
- B. 3 3/4
- C. 4 1/2
- D. 4 1/4

Answer: D

Explanation: Convert to fourths: 1 1/2 = 1 2/4; then 1 2/4 + 2 3/4 = 4 1/4.

Q029. **What is 3/5 of 40?**

- A. 24
- B. 8
- C. 15
- D. 32

Answer: A

Explanation: $3/5 \times 40 = 3 \times 8 = 24$.

Q030. **In a fraction, the denominator tells us—**

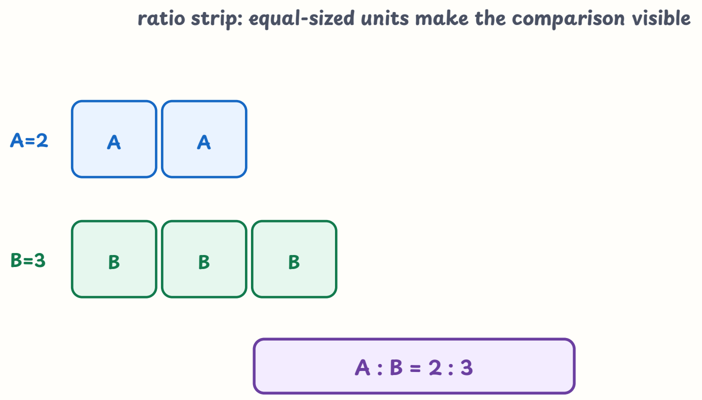
- A. the selected parts only
- B. the number of equal parts into which the whole is divided
- C. the value of the whole number only
- D. the remainder after division

Answer: B

Explanation: The denominator names the total number of equal parts; the numerator names selected parts.

Set 04 — Ratio, Proportion and Unitary Method (Q031–Q040)

Ratio compares quantities in the same units; proportion and unitary method extend the comparison to new values.



चित्र 3 — *equal unit strips show the ratio 2:3*

Q031. The ratio shown by the blue and green strips is—

- A. 3:2
- B. 2:5
- C. 2:3
- D. 5:3

Answer: C

Explanation: There are two equal blue units and three equal green units, so the ratio is 2:3.

Q032. The ratio 12:18 in simplest form is—

- A. 3:2
- B. 4:6
- C. 6:9
- D. 2:3

Answer: D

Explanation: Divide both terms by their HCF, 6: 12 ÷ 6 : 18 ÷ 6 = 2:3.

Q033. Which ratio is equivalent to 4:7?

- A. 12:21
- B. 8:11
- C. 16:24
- D. 20:28

Answer: A

Explanation: Multiplying both terms by 3 gives 12:21.

Q034. Six pens cost ₹48. At the same rate, eight pens cost—

- A. ₹56
- B. ₹64
- C. ₹72
- D. ₹96

Answer: B

Explanation: One pen costs ₹8; eight pens cost 8 × ₹8 = ₹64.

Q035. If the number of identical notebooks increases, their total cost increases in the same ratio. This is—

- A. inverse proportion
- B. no proportion
- C. direct proportion
- D. a constant difference

Answer: C

Explanation: When one quantity increases by a factor and the other does the same, the relation is direct proportion.

Q036. On a map, 1 cm represents 5 km. What distance does 7 cm represent?

- A. 12 km
- B. 30 km
- C. 40 km
- D. 35 km

Answer: D

Explanation: Using the scale, 7 × 5 = 35 km.

Q037. The ratio of girls to boys in a group is 3:2 and there are 25 learners. How many are girls?

- A. 15
- B. 10
- C. 12
- D. 20

Answer: A

Explanation: There are 5 total parts; each part is 25 ÷ 5 = 5, so girls = 3 × 5 = 15.

Q038. ₹360 is divided in the ratio 2:3. What is the smaller share?

- A. ₹120
- B. ₹144
- C. ₹180
- D. ₹216

Answer: B

Explanation: Total parts = 5; one part = ₹72, so the smaller share is 2 × ₹72 = ₹144.

Q039. If a:b = 4:5 and b:c = 10:3, then a:b:c is—

- A. 4:5:3
- B. 8:5:3
- C. 8:10:3
- D. 4:10:6

Answer: C

Explanation: Make the common term b equal to 10: 4:5 becomes 8:10, giving 8:10:3.

Q040. If 3/4 = x/20, then x equals—

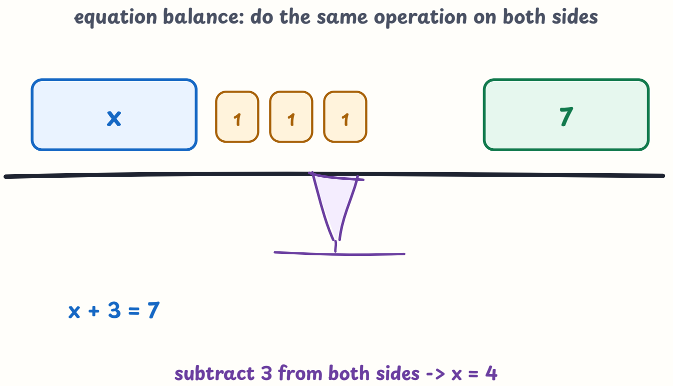
- A. 5
- B. 12
- C. 16
- D. 15

Answer: D

Explanation: Cross multiplication gives 4x = 60, so x = 15.

Set 05 — Algebraic Expressions, Equations and Patterns (Q041–Q050)

> Algebra uses symbols to express general relationships and unknown quantities.



चित्र 4 — both sides of an equation must remain balanced

Q041. The algebraic expression for "5 more than x" is—

- A. $x + 5$
- B. $5x$
- C. $x - 5$
- D. $x/5$

Answer: A

Explanation: Adding 5 to x represents five more than x.

Q042. Which pair consists of like terms?

- A. $3x$ and $5y$
- B. $3x$ and $-5x$
- C. x and x^2
- D. 4 and $4a$

Answer: B

Explanation: Like terms have the same variable part and exponent; $3x$ and $-5x$ are both x-terms.

Q043. If $a = 4$, what is the value of $2a + 3$?

- A. 8
- B. 10
- C. 11
- D. 14

Answer: C

Explanation: Substitute $a = 4$: $2(4) + 3 = 8 + 3 = 11$.

Q044. Solve: $x + 7 = 15$

- A. 7
- B. 9
- C. 22
- D. 8

Answer: D

Explanation: Subtract 7 from both sides: $x = 15 - 7 = 8$.

Q045. Solve: $3x = 27$

- A. 9
- B. 6
- C. 24
- D. 81

Answer: A

Explanation: Divide both sides by 3: $x = 27 \div 3 = 9$.

Q046. Solve: $2x + 5 = 17$

- A. 5
- B. 6
- C. 11
- D. 22

Answer: B

Explanation: Subtract 5 to get $2x = 12$, then divide by 2: $x = 6$.

Q047. The balance figure represents x + 3 = 7. What is the value of x?

- A. 3
- B. 7
- C. 4
- D. 10

Answer: C

Explanation: Removing three unit blocks from both sides leaves x = 7 − 3 = 4.

Q048. A rectangle has length x units and breadth 3 units. Its perimeter is—

- A. x + 3 units
- B. x + 6 units
- C. 2x + 3 units
- D. 2x + 6 units

Answer: D

Explanation: Perimeter = 2(length + breadth) = 2(x + 3) = 2x + 6.

Q049. The nth term of the pattern 3, 6, 9, 12, … is—

- A. 3n
- B. n + 3
- C. 3n + 1
- D. n^2

Answer: A

Explanation: The nth term is three times the position number: 3 × n.

Q050. If y = 2x + 1, what is y when x = 3?

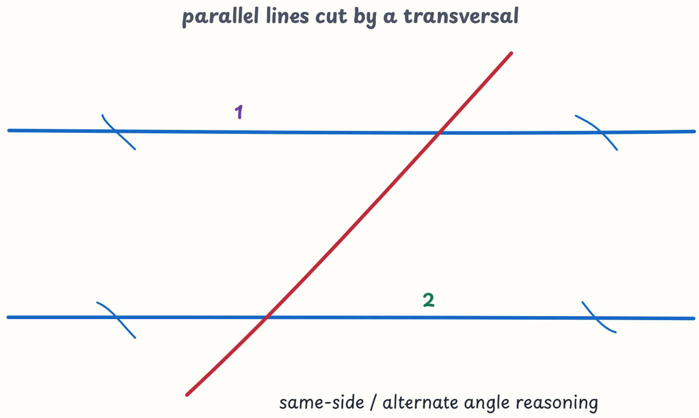
- A. 5
- B. 7
- C. 6
- D. 9

Answer: B

Explanation: Substitution gives y = 2(3) + 1 = 7.

Set 06 — Lines, Angles and Construction (Q051–Q060)

> Reasoning with angle relationships and construction steps is more important than memorising isolated names.



चित्र 5 — a transversal creates related angles on parallel lines

Q051. Two angles whose sum is 90° are called—

- A. supplementary angles
- B. vertically opposite angles
- C. complementary angles
- D. reflex angles

Answer: C

Explanation: Complementary angles together make a right angle of 90°.

Q052. Two angles whose sum is 180° are called—

- A. complementary angles
- B. acute angles
- C. vertical angles only
- D. supplementary angles

Answer: D

Explanation: Supplementary angles together make a straight angle of 180°.

Q053. Vertically opposite angles are always—

- A. equal
- B. supplementary only
- C. complementary only
- D. unequal

Answer: A

Explanation: When two lines intersect, opposite angles have equal measures.

Q054. **Two adjacent angles on a straight line are 110° and x°. What is x?**

- A. 20°
- B. 70°
- C. 110°
- D. 290°

Answer: B

Explanation: Angles on a straight line sum to 180°, so x = 180° − 110° = 70°.

Q055. **If one corresponding angle in the figure is 65°, the corresponding angle at the other parallel line is—**

- A. 25°
- B. 115°
- C. 65°
- D. 180°

Answer: C

Explanation: Corresponding angles are equal when a transversal cuts parallel lines.

Q056. **Two perpendicular lines form angles of—**

- A. 45° each
- B. 60° each
- C. 180° each
- D. 90° each

Answer: D

Explanation: Perpendicular lines meet at right angles, each measuring 90°.

Q057. **An angle bisector divides an angle into—**

- A. two equal angles
- B. three unequal angles
- C. two supplementary angles always
- D. one reflex angle

Answer: A

Explanation: By definition, an angle bisector creates two angles of equal measure.

Q058. To construct an equilateral triangle on a given segment, the essential instrument besides a ruler is a—

- A. protractor only
- B. compass
- C. set square only
- D. weighing balance

Answer: B

Explanation: Equal arcs from the two endpoints locate the third vertex; this requires a compass.

Q059. An angle measuring 180° is a—

- A. acute angle
- B. right angle
- C. straight angle
- D. reflex angle

Answer: C

Explanation: A straight angle forms a line and measures 180°.

Q060. If one pair of same-side interior angles is 65° in parallel lines, the other angle is—

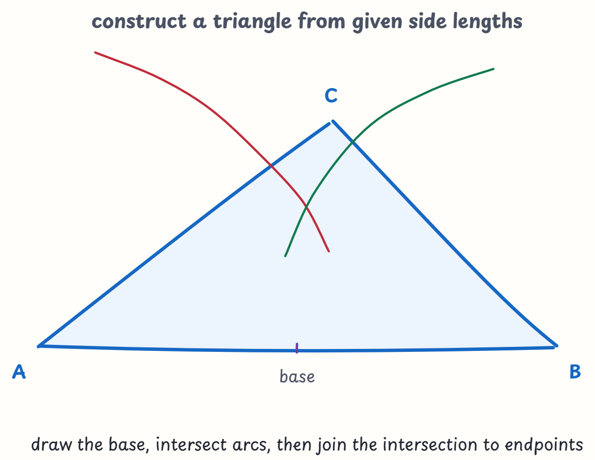
- A. 25°
- B. 65°
- C. 180°
- D. 115°

Answer: D

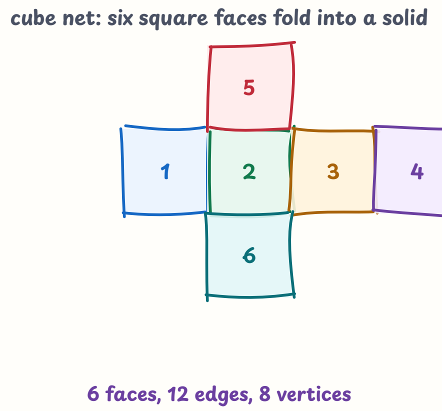
Explanation: Same-side interior angles are supplementary, so the other angle is 180° − 65° = 115°.

Set 07 — Triangles, Symmetry and Solids (Q061-Q070)

Triangles and solids should be connected with properties, visualisation and construction.



चित्र 6 — compass arcs locate the third vertex of a constructed triangle



चित्र 7 — six square faces form a cube net

Q061. The sum of the three interior angles of a triangle is—

- A. 180°
- B. 90°
- C. 270°
- D. 360°

Answer: A

Explanation: The angle-sum property of every triangle is 180°.

Q062. In the construction figure, after drawing the base and two arcs, the next step is to—

- A. erase the base
- B. join the arc intersection to both endpoints
- C. measure mass of the paper
- D. draw a circle through only one endpoint

Answer: B

Explanation: The arc intersection is the third vertex; joining it to the endpoints completes the triangle.

Q063. In a right-angled triangle, the side opposite the right angle is the—

- A. base angle
- B. shortest side always
- C. hypotenuse
- D. median only

Answer: C

Explanation: The hypotenuse is opposite the 90° angle and is the longest side of a right triangle.

Q064. Each angle of an equilateral triangle measures—

- A. 30°
- B. 45°
- C. 90°
- D. 60°

Answer: D

Explanation: All three angles are equal and their sum is 180°; each is 180° ÷ 3 = 60°.

Q065. A triangle with sides 5 cm, 5 cm and 8 cm is—

- A. isosceles
- B. equilateral
- C. scalene
- D. impossible because two sides are equal

Answer: A

Explanation: A triangle with two equal sides is isosceles.

Q066. How many lines of symmetry does a non-square rectangle have?

- A. 0
- B. 2
- C. 1
- D. 4

Answer: B

Explanation: A rectangle reflects onto itself across its horizontal and vertical mid-lines.

Q067. The order of rotational symmetry of an equilateral triangle is—

- A. 1
- B. 2
- C. 3
- D. 6

Answer: C

Explanation: A 120° turn, 240° turn and 360° turn each bring it onto itself, giving order 3.

Q068. How many faces does a cube have?

- A. 4
- B. 8
- C. 12
- D. 6

Answer: D

Explanation: A cube has six square faces, twelve edges and eight vertices.

Q069. How many vertices does a cuboid have?

- A. 8
- B. 6
- C. 10
- D. 12

Answer: A

Explanation: A cuboid has four vertices on each of two parallel rectangular faces, making 8.

Q070. The cube net shown above has how many square faces that fold to form a cube?

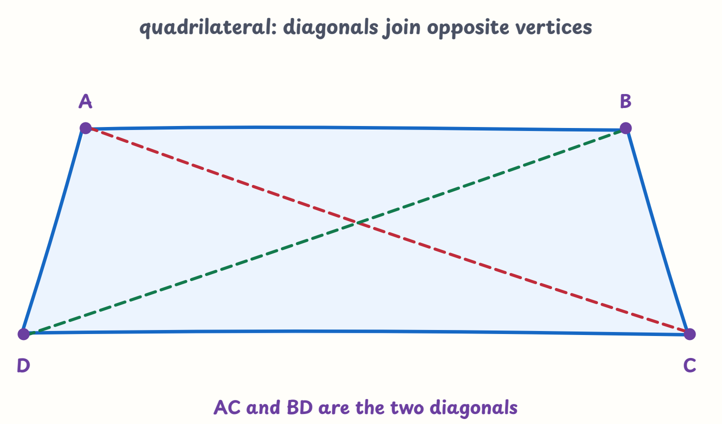
- A. 4
- B. 6
- C. 5
- D. 8

Answer: B

Explanation: A cube has six faces; a valid net contains six squares.

Set 08 — Quadrilaterals and Polygons (Q071-Q080)

Properties of quadrilaterals, diagonals and angle sums are useful for classification and reasoning.



चित्र 8 — AC and BD join opposite vertices of a quadrilateral

Q071. The sum of the interior angles of a quadrilateral is—

- A. 180°
- B. 270°
- C. 360°
- D. 540°

Answer: C

Explanation: A quadrilateral can be divided into two triangles, so the sum is $2 \times 180° = 360°$.

Q072. In a parallelogram, opposite sides are—

- A. perpendicular and unequal
- B. always curved
- C. equal only when it is a square
- D. parallel and equal

Answer: D

Explanation: Both pairs of opposite sides of a parallelogram are parallel and have equal lengths.

Q073. The diagonals of a rectangle are—

- A. equal in length
- B. always perpendicular
- C. unequal in every rectangle
- D. parallel to each other

Answer: A

Explanation: A rectangle has equal diagonals; perpendicular diagonals are a special property of some other quadrilaterals.

Q074. A quadrilateral with all four sides equal is a—

- A. trapezium only
- B. rhombus
- C. rectangle only
- D. kite only

Answer: B

Explanation: A rhombus is defined by four equal sides; a square is a special rhombus.

Q075. A square is both a—

- A. triangle and circle
- B. trapezium and triangle
- C. rectangle and rhombus
- D. kite only and not a rectangle

Answer: C

Explanation: A square has four right angles like a rectangle and four equal sides like a rhombus.

Q076. The diagonals of a parallelogram—

- A. are always equal
- B. never meet
- C. always meet at a right angle
- D. bisect each other

Answer: D

Explanation: Each diagonal cuts the other into two equal halves.

Q077. How many diagonals does a pentagon have?

- A. 5
- B. 3
- C. 6
- D. 10

Answer: A

Explanation: A polygon with n sides has $n(n-3)/2$ diagonals; for 5, this is 5.

Q078. The sum of the interior angles of a hexagon is—

- A. 360°
- B. 720°
- C. 540°
- D. 900°

Answer: B

Explanation: A hexagon is divided into 4 triangles, so its angle sum is $4 \times 180° = 720°$.

Q079. A quadrilateral with exactly one pair of parallel sides is commonly called a—

- A. rhombus
- B. rectangle
- C. trapezium
- D. square

Answer: C

Explanation: The defining property of a trapezium is one pair of parallel sides.

Q080. In the figure, AC and BD are—

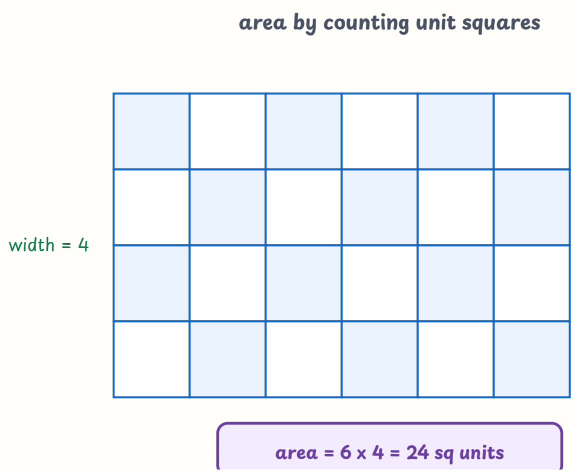
- A. adjacent sides
- B. parallel sides
- C. angle bisectors necessarily
- D. the diagonals of the quadrilateral

Answer: D

Explanation: A diagonal joins two opposite vertices; AC and BD do so.

Set 09 — Perimeter, Area and Volume (Q081-Q090)

Use units carefully and distinguish perimeter, area and volume.



चित्र 9 — count unit squares to connect area with length and breadth

Q081. The area represented by the grid is—

- A. 24 square units
- B. 10 square units
- C. 20 square units
- D. 48 square units

Answer: A

Explanation: There are 6 columns and 4 rows of unit squares: area = 6 × 4 = 24 square units.

Q082. A rectangle has length 12 cm and breadth 7 cm. Its perimeter is—

- A. 19 cm
- B. 38 cm
- C. 84 cm
- D. 42 cm

Answer: B

Explanation: Perimeter = 2(12 + 7) = 38 cm.

Q083. A square has area 64 cm^2. Its side is—

- A. 4 cm
- B. 16 cm
- C. 8 cm
- D. 32 cm

Answer: C

Explanation: $Side^2 = 64$, so side = $\sqrt{64}$ = 8 cm.

Q084. The area of a triangle with base 10 cm and height 6 cm is—

- A. 16 cm^2
- B. 60 cm^2
- C. 120 cm^2
- D. 30 cm^2

Answer: D

Explanation: Area = 1/2 × base × height = 1/2 × 10 × 6 = 30 cm^2.

Q085. The area of a parallelogram with base 9 cm and corresponding height 4 cm is—

- A. 36 cm^2
- B. 13 cm^2
- C. 18 cm^2
- D. 72 cm^2

Answer: A

Explanation: Area of a parallelogram = base × height = 9 × 4 = 36 cm^2.

Q086. The volume of a cuboid measuring 5 cm × 4 cm × 3 cm is—

- A. 12 cm^3
- B. 60 cm^3
- C. 47 cm^3
- D. 120 cm^3

Answer: B

Explanation: Volume = length × breadth × height = 5 × 4 × 3 = 60 cm^3.

Q087. If the side of a square is doubled, its area becomes—

- A. twice the original area
- B. three times the original area
- C. four times the original area
- D. half the original area

Answer: C

Explanation: Area is $side^2$; $(2s)^2 = 4s^2$.

Q088. 1 m^2 is equal to—

- A. 100 cm^2
- B. 1,000 cm^2
- C. 1,00,000 cm^2
- D. 10,000 cm^2

Answer: D

Explanation: Since 1 m = 100 cm, 1 m^2 = 100 × 100 = 10,000 cm^2.

Q089. The perimeter of a regular pentagon with each side 7 cm is—

- A. 35 cm
- B. 14 cm
- C. 28 cm
- D. 49 cm

Answer: A

Explanation: A pentagon has five equal sides, so perimeter = 5 × 7 = 35 cm.

Q090. A rectangle has area 96 cm^2 and length 12 cm. Its breadth is—

- A. 6 cm
- B. 8 cm
- C. 10 cm
- D. 14 cm

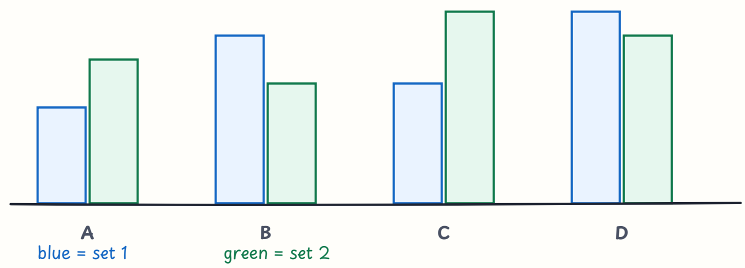
Answer: B

Explanation: Breadth = area ÷ length = 96 ÷ 12 = 8 cm.

Set 10 — Data Handling and Representation (Q091–Q100)

Read labels, scales and comparisons before performing calculations.

double bar graph: compare two data sets



चित्र 10 — compare two data sets across the same categories

Q091. According to the double bar graph, at category C the values for set 1 and set 2 are—

- A. 4 and 6
- B. 7 and 5
- C. 5 and 8
- D. 8 and 7

Answer: C

Explanation: The C bars show 5 for the first set and 8 for the second set.

Q092. What is the mean of 4, 7, 5 and 8?

- A. 5
- B. 7
- C. 24
- D. 6

Answer: D

Explanation: Mean = (4 + 7 + 5 + 8) ÷ 4 = 24 ÷ 4 = 6.

Q093. What is the median of 3, 5, 7, 9 and 11?

- A. 7
- B. 5
- C. 9
- D. 35

Answer: A

Explanation: The numbers are already ordered; the middle value is 7.

Q094. What is the mode of 2, 3, 3, 4 and 5?

- A. 2
- B. 3
- C. 4
- D. 17

Answer: B

Explanation: Mode is the value occurring most often; 3 occurs twice.

Q095. What is the range of 6, 9, 11 and 14?

- A. 5
- B. 6
- C. 8
- D. 20

Answer: C

Explanation: Range = maximum − minimum = 14 − 6 = 8.

Q096. A bar graph uses the scale 1 unit = 5 children. A bar of height 6 units represents—

- A. 11 children
- B. 25 children
- C. 60 children
- D. 30 children

Answer: D

Explanation: Multiply the scale value by the bar height: $6 \times 5 = 30$.

Q097. A class records the number of books read by four groups as 12, 15, 9 and 14. The total is—

- A. 50
- B. 41
- C. 46
- D. 56

Answer: A

Explanation: Adding the four frequencies gives $12 + 15 + 9 + 14 = 50$.

Q098. A graph that starts its vertical axis at 50 instead of 0 may—

- A. always show exact fractions
- B. make small differences look larger than they are
- C. remove the need for labels
- D. turn a bar graph into a pie chart

Answer: B

Explanation: A truncated scale can visually exaggerate differences, so axes and scales must be read critically.

Q099. Which display is most suitable for comparing two groups across the same categories?

- A. single-word list
- B. number line only
- C. double bar graph
- D. unlabelled picture

Answer: C

Explanation: A double bar graph places two bars per category for direct comparison.

Q100. When collecting data for a class survey, the first important step is to—

- A. draw bars before collecting any information
- B. choose the tallest bar in advance
- C. ignore who or what the data represent
- D. decide a clear question and identify what will be counted

Answer: D

Explanation: A clear question and defined variables make the collected data meaningful and interpretable.

Part 1 Quick Answer Index

001 A	002 B	003 C	004 D	005 A	006 B	007 C	008 D	009 A	010 B
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071 C	072 D	073 A	074 B	075 C	076 D	077 A	078 B	079 C	080 D
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Revision target: For content questions, draw a number line, strip, balance, grid or table whenever it clarifies the relationship. Always check units, signs, scale and whether a formula matches the shape.

CTET Paper II Mathematics MCQ

Part 2: Mathematics Pedagogy — Q101-Q200

Set 11 — Nature of Mathematics and Logical Thinking (Q101–Q110)

Paper-II mathematics pedagogy begins with the nature of mathematical thinking, reasoning, problem-solving and communication.

Q101. Mathematics is best understood in the classroom as—

- A. a way of reasoning, representing relationships, solving problems and communicating ideas
- B. only a collection of formulas
- C. only fast calculation without explanation
- D. a subject in which one method must always be copied

Answer: A

Explanation: Mathematics includes patterns, logic, representation and justification, not only numerical answers.

Q102. Which task best reflects the problem-solving nature of mathematics?

- A. Copy the area formula ten times
- B. Find different rectangles with area 24 cm^2 and explain how you know they work.
- C. Memorise a table without using it
- D. Underline every number in a paragraph

Answer: B

Explanation: The task involves exploring possibilities, applying a relationship and justifying the result.

Q103. Why should learners be asked to justify a mathematical answer?

- A. It makes every problem longer without purpose
- B. It tests handwriting more than mathematics
- C. Justification reveals the reasoning and helps distinguish a correct result from a lucky guess.
- D. It prevents learners from using representations

Answer: C

Explanation: Mathematical communication includes explaining why a method or conclusion is valid.

Q104. A learner notices that 2, 5, 8, 11 increases by 3. What should the teacher encourage next?

- A. Declare the pattern true for every possible number immediately
- B. Erase the sequence because it is not a formula
- C. Memorise only the first four terms
- D. Form a conjecture, test it with more terms and explain the pattern.

Answer: D

Explanation: A pattern can suggest a conjecture, but reasoning requires testing and explaining it.

Q105. In mathematics education, a child's error should first be treated as—

- A. evidence of the child's current reasoning that can guide instruction
- B. proof of fixed inability
- C. a reason for public punishment
- D. information that should always be ignored

Answer: A

Explanation: Error analysis reveals a strategy or misconception and helps the teacher plan the next step.

Q106. Which teacher response is most suitable for a learner showing mathematics anxiety?

- A. Give only timed tests until fear disappears
- B. Use supportive low-stakes tasks, allow explanation and build success gradually.
- C. compare the learner publicly with the fastest child
- D. remove the learner from all mathematical activity

Answer: B

Explanation: Safety, appropriate challenge and meaningful success help learners develop confidence and persistence.

Q107. Why are concrete and visual representations useful before a new abstract rule?

- A. They make symbols permanently unnecessary
- B. They are useful only for very young children
- C. They connect the symbol or rule with a meaning that learners can see, discuss and generalise.
- D. They replace reasoning with decoration

Answer: C

Explanation: Representations support conceptual understanding and can later be connected to formal notation.

Q108. A good mathematical problem should generally—

- A. have an answer that can be copied without thinking
- B. depend only on a memorised trick
- C. be unrelated to any mathematical idea
- D. invite learners to select strategies, make connections and reflect on the result

Answer: D

Explanation: Problem-solving involves understanding, strategy choice, execution and reflection.

Q109. When a learner asks "Why does this rule work?", the teacher should—

- A. explore the learner's question through examples, reasoning or a proof appropriate to the level
- B. say that rules never need reasons
- C. ask the learner to memorise the rule silently
- D. change the topic to avoid conceptual discussion

Answer: A

Explanation: Why-questions promote reasoning and help learners move beyond procedural memory.

Q110. The language of mathematics includes—

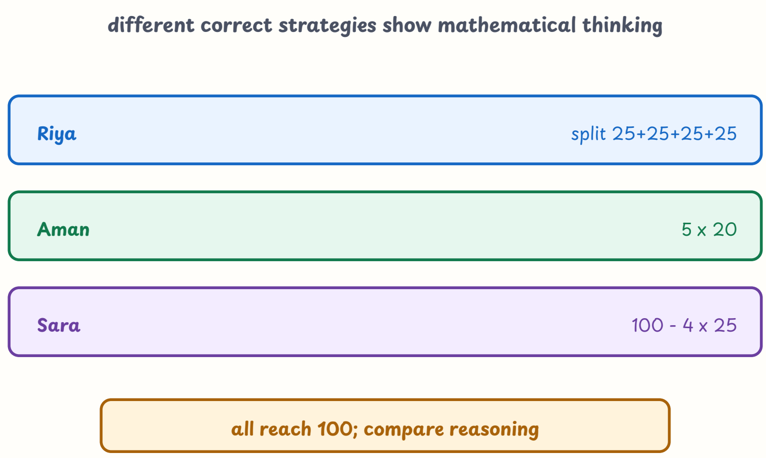
- A. only spoken English
- B. symbols, diagrams, words, tables and precise explanations
- C. only numerals with no interpretation
- D. only textbook definitions

Answer: B

Explanation: Mathematical ideas are communicated through multiple representations and precise relationships.

Set 12 — Children's Strategies and Constructivist Learning (Q111–Q120)

> A child's method is valuable evidence of thinking; the teacher's role is to elicit, compare and extend strategies.



चित्र 1 — different strategies can lead to the same correct result

Q111. When a child solves 48 + 27 by adding tens and ones separately, the teacher should—

- A. reject it because it is not the written algorithm
- B. give a harder question without listening
- C. ask the child to explain the strategy and connect it with place value
- D. tell the class that only one method is mathematical

Answer: C

Explanation: The explanation makes the child's place-value reasoning visible and allows comparison with other methods.

Q112. A non-standard method in mathematics is acceptable when it—

- A. is the fastest method in every problem
- B. uses no numbers or representations
- C. matches the teacher's method word for word
- D. is logically valid and the learner can communicate why it works

Answer: D

Explanation: Mathematical validity depends on reasoning, not on identical presentation.

Q113. Before teaching operations with integers, a teacher asks learners to discuss gains and losses. This uses—

- A. familiar experience as a bridge to a new mathematical idea
- B. an unrelated story to fill time
- C. assessment without any learning purpose
- D. rote learning instead of mathematical meaning

Answer: A

Explanation: Gains and losses provide a context for positive and negative change before formal notation.

Q114. A learner says 1/3 is greater than 1/2 because 3 is greater than 2. The teacher should first—

- A. mark the statement wrong and move on
- B. ask the learner to represent both fractions using equal wholes and discuss the comparison
- C. teach a rule about denominators without a model
- D. say that fractions cannot be compared

Answer: B

Explanation: An equal-whole visual model exposes why a larger denominator can mean smaller equal parts.

Q115. Why should learners compare two different solution strategies?

- A. They learn that only the longest method is correct
- B. They avoid explaining their own reasoning
- C. They can identify common mathematical ideas, efficiency and conditions under which a method works.
- D. They prove that answers do not matter

Answer: C

Explanation: Strategy comparison develops flexibility, connections and mathematical communication.

Q116. Using algebra tiles or counters while introducing an equation mainly helps learners—

- A. avoid all use of symbols permanently
- B. memorise answers without relationships
- C. learn only the names of colours
- D. represent unknowns and operations before connecting them to symbolic manipulation

Answer: D

Explanation: Manipulatives make the structure of an equation visible and can be linked to formal algebra.

Q117. Scaffolding in mathematics means that the teacher—

- A. provides temporary prompts, models or representations and gradually withdraws them
- B. solves every problem for the learner
- C. gives no help until the final test
- D. keeps learners on tasks they already master

Answer: A

Explanation: Temporary support helps learners perform beyond their independent level and move toward autonomy.

Q118. Which sequence gives multiple representations of the same ratio?

- A. four unrelated formulas
- B. verbal description, unit strips, table and symbolic ratio
- C. only a final answer
- D. a drawing with no labels or relationship

Answer: B

Explanation: Connecting representations helps learners see the same relationship in different forms.

Q119. In a constructivist mathematics classroom, knowledge is—

- A. transferred unchanged from teacher to learner
- B. developed only by repeating a definition
- C. actively constructed as learners connect experience, prior ideas, discussion and evidence
- D. independent of language and interaction

Answer: C

Explanation: Constructivist teaching treats learners as active meaning-makers and teachers as facilitators.

Q120. The strategy figure shows children reaching the same result in different ways. The best teacher action is to—

- A. select one child as intelligent and stop the discussion
- B. erase all methods except the teacher's
- C. award marks only for the longest written solution
- D. invite explanation, compare the methods and discuss why each is valid

Answer: D

Explanation: Comparing valid strategies develops reasoning and shows that mathematics can be represented flexibly.

Set 13 — Curriculum, Progression and Activity (Q121–Q130)

Curriculum should build connected concepts, mathematical habits and opportunities for meaningful activity.

Q121. A spiral approach to mathematics curriculum means—

- A. important ideas return at increasing levels of complexity and connection
- B. the same worksheet is repeated without change
- C. topics are taught only once
- D. learners move randomly between unrelated topics

Answer: A

Explanation: A spiral revisits ideas with greater depth as learners' understanding develops.

Q122. When planning a new lesson, the teacher should check whether it connects with —

- A. only the number of pages left
- B. learning goals, prior knowledge and the next conceptual step
- C. only the colour of the worksheet
- D. a topic unrelated to learners' current understanding

Answer: B

Explanation: Progression is meaningful when the new idea builds on what learners know and prepares later learning.

Q123. The concrete-representational-abstract sequence is useful because it—

- A. forbids abstract thinking
- B. requires every learner to use objects forever
- C. moves from meaningful objects or actions to pictures and then formal symbols
- D. replaces discussion with decoration

Answer: C

Explanation: The sequence supports a gradual connection between experience, representation and notation.

Q124. Using a local shop receipt to teach addition, discount and total cost illustrates—

- A. removing mathematics from daily life
- B. teaching only social studies
- C. assessing handwriting instead of calculation
- D. contextualising mathematics in a familiar real-life situation

Answer: D

Explanation: A familiar context can give mathematical operations purpose and support transfer.

Q125. The NCERT-linked Paper-II mathematics content should be taught mainly through—

- A. conceptual understanding, examples, problem-solving and classroom application
- B. memorisation of isolated answers
- C. only competitive speed drills
- D. copying every solved example without discussion

Answer: A

Explanation: CTET tests concepts, problem-solving and pedagogical understanding rather than recall alone.

Q126. A teacher links fractions, decimals and percentages in one sequence. This is valuable because—

- A. each topic becomes completely separate
- B. learners see equivalent representations and relationships across concepts
- C. it removes the need for examples
- D. it guarantees that every learner uses one method

Answer: B

Explanation: Connected concepts help learners transfer understanding between forms.

Q127. A mathematics learning objective is strongest when it describes—

- A. only the chapter title
- B. only the teacher's activity
- C. what learners will understand or be able to do and how evidence will show it
- D. the number of minutes without a learning purpose

Answer: C

Explanation: Clear objectives align instruction, activity and assessment.

Q128. Which classroom activity best develops mathematical reasoning?

- A. Copy the names of shapes ten times
- B. colour every shape the same colour
- C. memorise one definition without examining examples
- D. Sort several quadrilaterals in more than one way and justify each grouping.

Answer: D

Explanation: Multiple classifications require learners to identify properties and explain their decisions.

Q129. Mental mathematics is most educationally useful when learners—

- A. use flexible strategies, estimate, explain and check whether an answer is reasonable
- B. race for an answer without understanding
- C. avoid all written representations
- D. memorise only one shortcut for every problem

Answer: A

Explanation: Mental calculation builds number sense when reasoning and estimation accompany fluency.

Q130. The community-mathematics figure can best support a lesson by asking learners to—

- A. memorise the labels in the figure
- B. collect and solve real problems involving money, time, measure or shape in their surroundings
- C. avoid using any mathematical language
- D. replace all classroom concepts with unexamined guesses

Answer: B

Explanation: Community contexts make mathematics meaningful while still requiring defined quantities and reasoning.

Set 14 — Language of Mathematics and Classroom Discourse (Q131–Q140)

Language can either support or obstruct mathematical meaning; teachers should make terms, symbols and explanations accessible.

Q131. Why should a teacher distinguish the everyday and mathematical meanings of words such as "difference" or "volume"?

- A. Everyday meanings are always wrong
- B. Mathematical words never need context
- C. The distinction prevents language confusion and helps learners interpret mathematical tasks precisely.
- D. Definitions alone guarantee problem-solving

Answer: C

Explanation: Some words have specialised mathematical meanings; discussing context supports comprehension.

Q132. A learner can calculate but cannot explain the steps. The teacher should encourage—

- A. only faster calculation
- B. silence during problem-solving
- C. copying a model answer without meaning
- D. oral, written or diagrammatic explanation using mathematical vocabulary

Answer: D

Explanation: Communication reveals reasoning and strengthens links between operations, representations and language.

Q133. A bilingual word wall containing "factor", "multiple" and their home-language equivalents is—

- A. a multilingual resource that can make mathematical vocabulary accessible
- B. a barrier that prevents English learning
- C. useful only for decoration
- D. an assessment that should replace instruction

Answer: A

Explanation: Home-language connections can support concept formation while learners acquire the school language.

Q134. Before solving a word problem, learners should be helped to—

- A. circle every number and multiply them automatically
- B. identify the quantities, relationships and question represented by the language
- C. ignore the context and choose a familiar formula
- D. translate each word without considering the relationship

Answer: B

Explanation: Understanding the situation is necessary before selecting an operation or representation.

Q135. Which prompt best develops mathematical reading?

- A. How many letters are in the symbol's name?
- B. Copy the symbol on every line
- C. What does the symbol $\leq$ tell us in this statement, and which values satisfy it?
- D. Read the statement aloud without interpreting it

Answer: C

Explanation: The prompt asks learners to interpret notation and connect it with a set of values.

Q136. A learner interprets "at least 6" as exactly 6. A good response is to—

- A. declare the learner careless
- B. avoid all language in inequalities
- C. teach only a symbol with no examples
- D. use number-line examples and contexts to compare "at least", "exactly" and "at most"

Answer: D

Explanation: Contrasting contexts and representations clarify the meaning of inequality phrases.

Q137. If a learner struggles with the language of a mathematics problem, the teacher should —

- A. read it with the learner, use a diagram or simpler wording and then return to the mathematical relationship
- B. assume the learner lacks mathematical ability
- C. remove all word problems permanently
- D. give the numerical answer before discussing the situation

Answer: A

Explanation: Language support should remove an access barrier without lowering the mathematical goal.

Q138. Which student response gives the strongest evidence of mathematical understanding?

- A. The learner writes only a final number
- B. The learner gives an answer, represents it in a diagram and explains why the method works.
- C. The learner copies the formula but cannot use it
- D. The learner repeats a definition unrelated to the task

Answer: B

Explanation: Understanding is better evidenced by connected representation, reasoning and conclusion.

Q139. Mathematical classroom discourse is productive when learners—

- A. speak only to repeat the teacher
- B. compete to finish without explaining
- C. listen, question, compare methods and use evidence to agree or disagree respectfully
- D. avoid alternative strategies

Answer: C

Explanation: Discussion helps learners make reasoning public and refine mathematical ideas.

Q140. The teacher asks, "How do you know your answer is reasonable?" This mainly promotes—

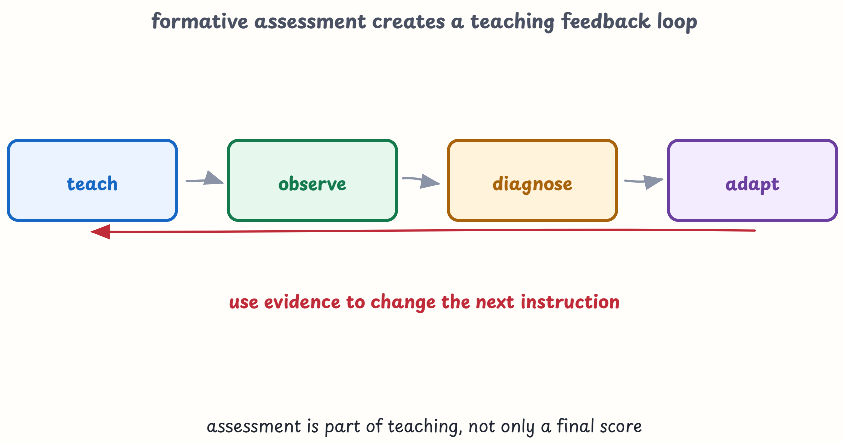
- A. rote recall of a definition
- B. silent copying of an algorithm
- C. speed without checking
- D. estimation, justification and reflection on the result

Answer: D

Explanation: Reasonableness questions connect calculation with estimation and mathematical judgement.

Set 15 — Evaluation and Assessment in Mathematics (Q141–Q150)

> Assessment should reveal mathematical thinking and inform the next teaching decision.



चित्र 2 — use mathematical evidence to adapt the next teaching step

Q141. A teacher uses a quick question during a lesson and changes the next example after seeing responses. This is—

- A. formative assessment
- B. only a final examination
- C. a permanent ability label
- D. summative certification

Answer: A

Explanation: The evidence is collected during learning and used immediately to adapt teaching.

Q142. Diagnostic assessment in mathematics is intended to identify—

- A. only the class rank
- B. a specific misconception, strategy or prerequisite gap
- C. the learner's permanent intelligence
- D. which child writes most neatly

Answer: B

Explanation: Diagnosis should explain the nature of a difficulty so that support can be targeted.

Q143. A term-end mathematics test used to report achievement is mainly—

- A. diagnostic assessment before teaching
- B. informal peer discussion
- C. summative assessment
- D. a learning game without evidence

Answer: C

Explanation: Summative assessment summarises achievement at the end of a period.

Q144. A mathematics assessment has validity when it—

- A. contains the hardest possible numbers
- B. gives identical marks to every learner
- C. uses only lengthy questions
- D. measures the intended mathematical understanding rather than irrelevant reading or handwriting difficulty

Answer: D

Explanation: Validity concerns whether evidence supports the intended interpretation of learning.

Q145. Reliability of a mathematics scoring process means—

- A. similar performances receive reasonably consistent scores under comparable conditions
- B. every task has only one possible method
- C. the worksheet has colourful borders
- D. the test is always very difficult

Answer: A

Explanation: Reliable scoring is consistent, while validity concerns whether the right construct is measured.

Q146. A rubric for a mathematical investigation may include—

- A. only final-answer speed
- B. understanding, strategy, representation, reasoning and communication
- C. only handwriting size
- D. only the number of pages used

Answer: B

Explanation: A rubric makes multiple dimensions of mathematical performance visible.

Q147. An observation checklist during group problem-solving should record—

- A. a general label such as clever
- B. family background
- C. specific behaviours such as explaining, representing, listening and revising
- D. only who speaks loudest

Answer: C

Explanation: Observable criteria provide useful evidence without reducing a learner to a vague label.

Q148. When a teacher studies several wrong solutions to the same problem, the purpose is to—

- A. prove that learners should not attempt problems
- B. choose the neatest handwriting
- C. replace teaching with ranking
- D. infer common strategies or misconceptions and plan responsive instruction

Answer: D

Explanation: Patterns in errors reveal thinking and guide the next learning experience.

Q149. Assessment for learning in mathematics is strongest when learners—

- A. receive specific feedback and use it to revise a method or explanation
- B. see only a final rank
- C. are given marks without criteria
- D. repeat an identical test without feedback

Answer: A

Explanation: Feedback becomes formative when it informs the learner's next action.

Q150. The assessment-cycle figure represents which sequence?

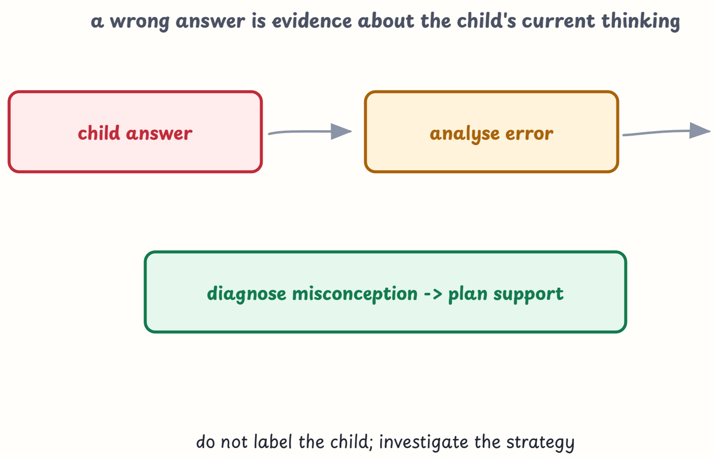
- A. test, rank, punish and stop
- B. teach, collect evidence, diagnose and adapt the next instruction
- C. memorise, copy, erase and repeat
- D. grade once and ignore the evidence

Answer: B

Explanation: Formative assessment is a feedback loop connecting evidence with teaching decisions.

Set 16 — Problems of Teaching and Mathematical Errors (Q151–Q160)

A problem in performance may arise from concepts, language, representations, instruction or confidence; investigate before labelling.



चित्र 3 — trace a wrong answer to the child's strategy, then plan support

Q151. Several learners write 402 as 42. What should the teacher investigate first?

- A. Whether they can memorise more number names
- B. Whether to give them harder numbers
- C. Their understanding of zero as a place-holder and their interpretation of place value.
- D. Whether to stop using written numerals

Answer: C

Explanation: The error suggests a place-value or zero-placeholder issue that needs evidence and targeted representation.

Q152. A learner has persistent difficulty understanding number combinations despite appropriate instruction. The teacher should—

- A. immediately label the learner lazy
- B. remove all mathematics from the timetable
- C. compare the learner publicly with peers
- D. observe patterns, provide targeted support and collaborate with family or specialists when needed

Answer: D

Explanation: Persistent difficulty needs careful observation and support; it is not proof of laziness or low intelligence.

Q153. Which response is most likely to reduce mathematics anxiety?

- A. Allow think time, multiple representations, supportive feedback and opportunities to revise.
- B. Use public speed contests as the main assessment
- C. penalise every question asked
- D. give unpredictable tasks without explaining the goal

Answer: A

Explanation: Predictability, support and a growth-oriented classroom make mathematical participation safer.

Q154. A learner solves a problem orally but struggles with the dense wording of the written version. The teacher should—

- A. conclude that the learner cannot do mathematics
- B. unpack the language and use a diagram while preserving the mathematical challenge
- C. give only the final answer
- D. remove all written mathematics permanently

Answer: B

Explanation: Language scaffolding can separate a reading barrier from the mathematical reasoning being assessed.

Q155. A learner places −5 to the right of −2. Which representation is likely to help?

- A. a multiplication table only
- B. a list of unrelated formulas
- C. A number line with movement and comparison of positions
- D. a geometry net

Answer: C

Explanation: The number line makes order and direction of integers visible.

Q156. A child adds fractions by adding both numerators and denominators. The teacher should—

- A. teach the cross rule without any meaning
- B. say the child must never use pictures
- C. mark every future fraction answer zero
- D. use equal-part models to discuss why denominators describe the unit and then connect to like denominators

Answer: D

Explanation: The misconception concerns the meaning of the denominator; models can support reconstruction of the operation.

Q157. A learner calls every four-sided figure a square. The best first step is to—

- A. sort and compare examples by properties such as equal sides and right angles
- B. ask the learner to memorise a longer definition
- C. avoid showing non-square quadrilaterals
- D. accept the word without examining figures

Answer: A

Explanation: Classification by properties helps learners distinguish a square from the broader class of quadrilaterals.

Q158. If learners can apply a formula only to the exact example shown, the teacher should provide —

- A. more copies of the same example
- B. varied contexts and representations that require selecting and adapting the relationship
- C. only a longer formula list
- D. answers before learners analyse the problem

Answer: B

Explanation: Transfer develops when learners use concepts across varied but related situations.

Q159. Remedial mathematics should be—

- A. the same extra worksheet for everyone
- B. a punishment for low marks
- C. specific to the diagnosed need, conceptually meaningful and followed by reassessment
- D. only repetitive copying of answers

Answer: C

Explanation: Effective remediation matches a learner's difficulty and checks whether the support helped.

Q160. The error-analysis figure suggests that a wrong answer should lead to—

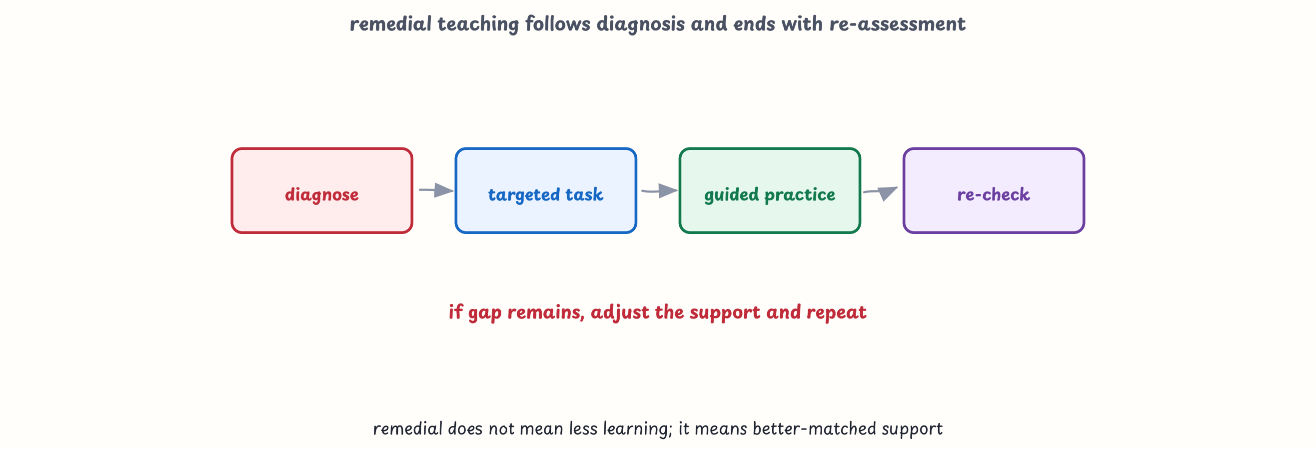
- A. immediate punishment
- B. a fixed label
- C. ignoring all future evidence
- D. analysis of the strategy, diagnosis of the misconception and a planned response

Answer: D

Explanation: The figure models an evidence-based cycle rather than a judgement about the child.

Set 17 — Diagnostic and Remedial Teaching (Q161–Q170)

Remediation is a cycle: find the gap, design a matched experience, observe progress and adjust.



चित्र 4 — diagnose, target support, practise, re-check and adjust

Q161. What is the first step in planning remedial mathematics teaching?

- A. Diagnose the learner's particular misconception or prerequisite gap.
- B. Give a random extra worksheet
- C. announce a lower expectation
- D. repeat the same test without analysis

Answer: A

Explanation: Without diagnosis, support may not address the actual source of difficulty.

Q162. A learner confuses perimeter and area. Which remedial activity is most appropriate?

- A. Memorise both formulas ten times
- B. Cover shapes with unit squares and trace their boundary, then compare the two measures.
- C. solve only volume questions
- D. copy a definition without measuring anything

Answer: B

Explanation: The activity contrasts inside coverage with boundary length and connects each to its unit.

Q163. After a remedial activity, the teacher should—

- A. assume the gap is closed without evidence
- B. give a permanent label
- C. use a fresh but related task to check transfer and decide the next support
- D. stop teaching the concept entirely

Answer: C

Explanation: Reassessment shows whether understanding transfers beyond the original practice.

Q164. Which grouping is most useful for targeted remedial teaching?

- A. A permanent group based on one old test
- B. a group based only on seating order
- C. a group that excludes peer explanation
- D. A flexible group formed from a shared diagnosed need and reviewed as learners progress.

Answer: D

Explanation: Flexible grouping aligns support with current evidence and avoids fixed labels.

Q165. Why is giving more of the same difficult worksheet often poor remediation?

- A. It may repeat the obstacle without changing the representation, explanation or strategy.
- B. Practice can never be useful
- C. All learners need exactly one task
- D. Difficulty is always caused by laziness

Answer: A

Explanation: Remediation should change the support and address the diagnosed need, not merely increase quantity.

Q166. A short interview asking "How did you get this answer?" is useful because it—

- A. measures handwriting speed
- B. reveals the learner's reasoning, not only the correctness of the final answer
- C. replaces every form of assessment
- D. guarantees an immediate correct answer

Answer: B

Explanation: Learner explanations provide diagnostic evidence about strategy and misconception.

Q167. For a place-value gap, which sequence is most appropriate?

- A. start with large algebraic equations
- B. memorise number names without quantities
- C. bundle or represent units, tens and hundreds, record a place-value table, then use numerals
- D. avoid exchanging or regrouping models

Answer: C

Explanation: Concrete and tabular representations build meaning before formal notation.

Q168. For a ratio misconception, the teacher should first use—

- A. a percentage formula with no context
- B. a random set of unrelated numbers
- C. only a final ratio answer
- D. equal unit strips or a table showing how both quantities scale together

Answer: D

Explanation: Unit strips and tables make the multiplicative relationship visible.

Q169. A learner improves after guided practice but still needs prompts. The teacher should—

- A. continue appropriate support, fade prompts gradually and provide independent checks
- B. remove all support suddenly
- C. keep the learner dependent forever
- D. declare the learner fully remedied without checking

Answer: A

Explanation: Fading support balances access with development of independence.

Q170. The remedial-cycle figure ends with re-checking because—

- A. testing is more important than teaching
- B. new evidence shows whether the intervention worked and what should happen next
- C. every learner must repeat the same test
- D. remedial teaching ends before learning is observed

Answer: B

Explanation: Re-checking makes remediation responsive and cyclical.

Set 18 — Inclusive and Differentiated Mathematics Classroom (Q171–Q180)

Inclusion means removing barriers while keeping worthwhile mathematical participation and expectations.

Q171. Universal Design for Learning in mathematics encourages—

- A. one fixed explanation and one response mode
- B. separate goals for every learner without shared participation
- C. multiple ways to access a concept, engage with a task and show understanding
- D. removing challenging mathematics from the curriculum

Answer: C

Explanation: UDL anticipates learner variability and offers flexible routes to the same meaningful goal.

Q172. How can a teacher make a geometry lesson more accessible to a learner with visual impairment?

- A. exclude the learner from construction activities
- B. use only tiny printed diagrams
- C. assess only visual copying
- D. Use tactile shapes, raised diagrams, verbal descriptions and an accessible way to respond.

Answer: D

Explanation: Tactile and verbal representations provide access to geometric properties and reasoning.

Q173. During a mathematics discussion, support for a learner with hearing difficulty may include—

- A. written instructions, visible teacher positioning, captions or a peer-supported visual record
- B. speaking while facing away with no written support
- C. removing the learner from group work
- D. assuming that louder speech solves every barrier

Answer: A

Explanation: Accessible visual information and positioning support participation without lowering the mathematical goal.

Q174. An advanced learner finishes routine work quickly. The best response is to—

- A. give only more copies of the same sums
- B. offer a challenging investigation requiring generalisation, proof or multiple representations
- C. make the learner teach every lesson
- D. allow the learner to skip all mathematical reasoning

Answer: B

Explanation: Enrichment deepens mathematical thinking instead of adding repetitive quantity.

Q175. In a mixed-ability group, the teacher should—

- A. let the highest scorer do all calculations
- B. give the same fixed role to one learner
- C. assign rotating roles and use tasks with entry points and extensions so all can reason
- D. separate learners permanently by marks

Answer: C

Explanation: Flexible roles and accessible challenge promote participation and peer learning.

Q176. A gender-sensitive mathematics classroom should—

- A. assign equipment only to boys
- B. expect girls to copy while boys explain
- C. avoid asking girls conceptual questions
- D. give learners of all genders equal access to manipulatives, leadership and challenging tasks

Answer: D

Explanation: Equity in participation challenges stereotypes and expands opportunity.

Q177. A learner uses a home-language term while explaining a mathematical idea. The teacher should—

- A. value the explanation, connect the term to school-language vocabulary and continue the reasoning
- B. reject the idea before understanding it
- C. ban all home-language support
- D. assume the learner has no mathematical concept

Answer: A

Explanation: Language bridging preserves mathematical thinking while developing the language of instruction.

Q178. For a learner from a resource-constrained background, an equitable mathematics response is to—

- A. assume the learner cannot handle concepts
- B. provide school-based materials, peer support and meaningful practice rather than lower expectations
- C. require costly materials from home
- D. give only trivial work

Answer: B

Explanation: Equity supplies needed access while maintaining worthwhile learning goals.

Q179. A learner with attention-regulation difficulty may benefit from—

- A. long unbroken lectures only
- B. public punishment for every movement
- C. short structured steps, visible goals, movement breaks and feedback on progress
- D. removal from all problem-solving

Answer: C

Explanation: Structure and manageable chunks can support sustained attention and participation.

Q180. Success in an inclusive group mathematics task should be judged mainly by —

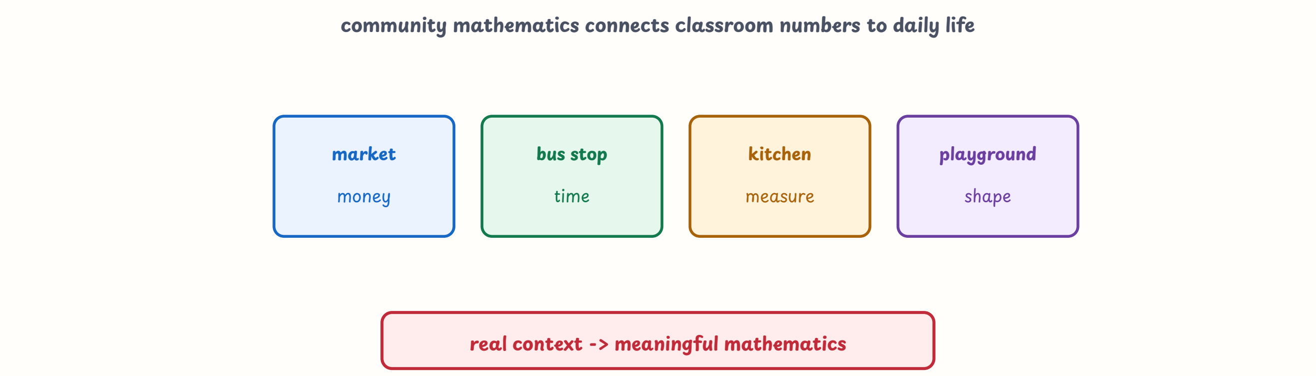
- A. which learner finishes first
- B. how similar every notebook looks
- C. whether the group has one silent leader
- D. mathematical learning, reasoning and meaningful participation with appropriate support

Answer: D

Explanation: Inclusive success values both mathematical evidence and access to participation.

Set 19 — Community Mathematics and Real-Life Application (Q181–Q190)

> Community mathematics links formal ideas with authentic situations while keeping quantities, assumptions and reasoning explicit.



चित्र 5 — market, transport, kitchen and playground contexts can generate mathematics

Q181. A teacher uses a market bill to teach total, change and comparison. The main pedagogical benefit is—

- A. learners use mathematics for a purposeful everyday situation
- B. learners avoid all formal notation
- C. market language automatically teaches every concept
- D. the teacher no longer needs to assess reasoning

Answer: A

Explanation: Real contexts create purpose, but learners still need to calculate, represent and justify.

Q182. Which community-data activity best supports data handling?

- A. copy a ready-made graph without reading it
- B. Survey a local travel pattern, organise responses in a table and discuss a suitable graph.
- C. choose a graph before asking any question
- D. collect unlabelled numbers with no variable

Answer: B

Explanation: A meaningful survey connects question, data collection, representation and interpretation.

Q183. Measuring a classroom or playground can help learners connect—

- A. only multiplication facts
- B. mathematics with no physical meaning
- C. units, estimation, instruments, perimeter, area and the accuracy of measurement
- D. a formula with no need for units

Answer: C

Explanation: Measurement in a real space supports estimation, unit sense and geometric application.

Q184. A class plans a small event budget. This can develop—

- A. only decorative writing
- B. a single memorised algorithm
- C. mathematics without numbers
- D. operations, estimation, ratio, decision-making and justification of choices

Answer: D

Explanation: Budget decisions require quantities, comparisons and explanation of assumptions.

Q185. When using a cultural or local context in mathematics, the teacher should—

- A. invite varied experiences without assuming that one community practice represents everyone
- B. treat one family's practice as universal
- C. avoid all local knowledge
- D. use culture only as decoration

Answer: A

Explanation: Culturally responsive teaching values diversity and checks assumptions.

Q186. A community-based mathematics project should include—

- A. only photographs with no mathematical record
- B. a clear question, defined quantities, data or measurements, analysis and communication
- C. an answer chosen before investigation
- D. unverified claims without units

Answer: B

Explanation: A project becomes mathematical when the situation is represented and reasoned about systematically.

Q187. Which activity best develops mathematical modelling?

- A. copy a formula without a situation
- B. give a number with no assumptions
- C. Represent water use with variables or a table, make assumptions, calculate and discuss limits of the model.
- D. avoid checking whether the model fits reality

Answer: C

Explanation: Modelling translates a real situation into mathematics and examines how well the representation works.

Q188. If learners compare prices from different shops, the teacher should also ask them to consider—

- A. only the largest printed number
- B. the colour of each shop sign
- C. a price without identifying what it measures
- D. units, quantity, quality and whether the comparison is fair

Answer: D

Explanation: Fair comparison requires common units and attention to what each number represents.

Q189. Why should a teacher not assume that every learner has the same experience of a bus timetable or market?

- A. Learners' access and experiences differ, so the context should be explained and alternatives offered.
- B. Real-life contexts are never useful
- C. only textbook contexts are inclusive
- D. mathematical meaning depends on one family's experience

Answer: A

Explanation: Inclusive contextualisation provides background and multiple entry points.

Q190. The community-mathematics figure is best used as—

- A. a poster whose labels must be memorised
- B. a prompt for learners to generate and solve their own measurable questions
- C. a replacement for mathematical discussion
- D. proof that one context fits every learner

Answer: B

Explanation: Learner-generated questions connect context, agency and mathematical reasoning.

Set 20 — Classroom Scenarios and Integrated Revision (Q191–Q200)

These scenarios combine content, child thinking, assessment and the most appropriate teacher response.

Q191. A learner says 1/3 > 1/2 because 3 > 2. Which teacher response is most appropriate?

- A. Say that the denominator rule is simply the opposite
- B. mark the answer wrong without discussion
- C. Use equal wholes or fraction strips, ask the learner to compare unit sizes and explain the conclusion.
- D. avoid fractions until the learner memorises tables

Answer: C

Explanation: The error comes from comparing denominators without considering the size of equal parts; a visual model supports reconstruction.

Q192. A learner orders −4, −1 and −6 as −6, −1, −4. What should the teacher do?

- A. teach multiplication signs instead
- B. ask the learner to copy the order fifty times
- C. say negative numbers cannot be ordered
- D. Place the integers on a number line and discuss left-to-right order and distance from zero.

Answer: D

Explanation: The number line gives a visual meaning to integer order and corrects the direction misconception.

Q193. While solving 2x + 3 = 11, a learner removes 3 from only one side. The best response is to—

- A. use a balance model and ask what operation keeps both sides equal
- B. give the answer x = 4 without explanation
- C. say equations are just tricks
- D. remove all algebra symbols

Answer: A

Explanation: The balance representation makes equality and the need for the same operation on both sides explicit.

Q194. A learner scales 2:3 to 4:5. How should the teacher respond?

- A. say that any larger numbers make an equivalent ratio
- B. Use unit strips or a ratio table to show that both terms must be multiplied by the same factor.
- C. teach only cross multiplication without interpretation
- D. accept the answer because both terms increased

Answer: B

Explanation: Equivalent ratios preserve the multiplicative relationship between both quantities.

Q195. A learner finds the area of a rectangle by adding length and breadth. The teacher should—

- A. give only the formula to memorise
- B. say area and perimeter are identical
- C. cover or draw unit squares and contrast the inside area with the boundary perimeter
- D. avoid measuring rectangles

Answer: C

Explanation: Unit squares make area as covering visible and distinguish it from boundary length.

Q196. A learner reads a bar graph correctly but ignores a scale of 1 unit = 10. The teacher should—

- A. mark all graph work permanently wrong
- B. remove scales from future graphs
- C. ask only for the tallest bar
- D. ask the learner to label the scale and interpret one bar using the scale

Answer: D

Explanation: The difficulty is interpreting the scale; a focused representation and explanation target that need.

Q197. A teacher notices that a timed test produces very low scores, but discussion shows sound reasoning. The teacher should—

- A. use more than one form of evidence and examine whether speed is obscuring the intended understanding
- B. conclude that discussion is not mathematical
- C. lower every learning goal
- D. use only timed tests from then on

Answer: A

Explanation: Fair evaluation aligns evidence with the intended construct and recognises mathematical reasoning.

Q198. A teacher has shown one algorithm for every problem. To develop flexible thinking, the teacher should—

- A. forbid all other methods
- B. invite alternative methods, compare their reasoning and discuss when each is efficient
- C. give only harder versions of the same algorithm
- D. assess handwriting instead of strategy

Answer: B

Explanation: Multiple valid strategies develop flexibility, communication and deeper understanding.

Q199. Learners complete an exit ticket and identify the step they found difficult. The teacher can use it to—

- A. assign a permanent ability label
- B. replace all future teaching with tests
- C. plan the next example or flexible group and invite learners to revise their reasoning
- D. ignore the responses after recording marks

Answer: C

Explanation: Learner reflection plus teacher evidence supports formative planning and metacognition.

Q200. Which teacher best reflects CTET Paper-II Mathematics pedagogy?

- A. A teacher who values only speed and one fixed algorithm
- B. A teacher who labels errors as inability and avoids discussion
- C. A teacher who completes the textbook without checking understanding
- D. A teacher who connects concepts with representations and life, elicits strategies, analyses errors, includes every learner and uses assessment to adapt teaching.

Answer: D

Explanation: The strongest pedagogy integrates conceptual understanding, child thinking, inclusion, communication, assessment and remedial action.

Part 2 Quick Answer Index

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Revision target: For Paper-II pedagogy, practise reading the child's method before choosing the teacher response. The correct option usually makes thinking visible, uses an appropriate representation, supports participation and turns assessment into the next teaching step.